

	ITER, SIKSHA 'O' ANUSANDHAN (Deemed to be University)		Assignment
Branch	All	Programme	B.Tech
Course Name	Science and Technology in Ancient India	Semester	VIII
Course Code	IKS4003	Academic Year	2026/Even
Assignment-3	Topic- Indian response to new scientific knowledge	GP-1	
Learning Level (LL)	L1: Remembering	L3: Applying	L5: Evaluating
	L2: Understanding	L4: Analyzing	L6: Creating
Q's	Questions		COs
1	Mention two pioneering products introduced by CSIR that had a significant impact on India's food and agricultural sectors.		CO1
2	Which early Indian observatory, established between 1780 and 1790, initiated studies in climatology, meteorology, and weather prediction?		CO1
3	What were the primary objectives behind the establishment of the Indian Space Research Organisation (ISRO)?		CO1
4	List any two major services or applications provided by the Indian National Satellite (INSAT) system.		CO1
5	Who designed India's three-stage nuclear program, and what specific natural resources was it designed to harness?		CO1
6	According to ancient Indian cosmology, what are the three realms of existence?		CO1
7	How did ancient Indians visualize the spatial relationship between the earth, sky, and air?		CO1
8	Name two sacred rivers and two major mountain ranges that prove ancient Indians had accurate knowledge of physical geography.		CO1
9.	Name two foreign observers who documented the geography of India for the outside world.		CO2
10	What does the term "Seven Dvipas" refer to in ancient Indian texts like the Puranas, and what was the role of Jambudvipa?		CO2

11	How was the territorial extent of "Bharatavarsha" described in ancient texts?	CO1	L18
12	Ancient Indian geography classified Bharatavarsha into five main regional divisions. Name any four of these divisions along with their geographic direction.	CO2	L20
13	What specific natural features did the Puranas document that make them a comprehensive source of ancient geography?	CO4	L21
14	Define archaeological chemistry and mention one of its key applications in the study of ancient technologies.	CO2	L22
15	According to Kautilya's Arthashastra, what administrative positions were created to oversee metallurgy and mining during the Mauryan period? Name any two.	CO5	L23
16	What classical account by Herodotus provides early literary evidence of the use of iron by ancient Indians?	CO2	L24
17	Why are excavated metal coins often preferred over other miscellaneous metal artefacts for scientific examination in archaeology?	CO3	L25
18	Based on scientific studies, what evidence suggests that the people of the Painted Grey Ware (P.G.W.) culture were adept at ceramic coloring techniques?	CO3	L26
19	What was the approximate maximum temperature that ancient Indians could achieve in their high-temperature furnaces during the early historic phase?	CO3	L27
20	What method did Acharya Brajendranath Seal recommend for studying the scientific elements in ancient Indian texts?	CO2	L28
21	How did the ancient scholar Caraka describe the function of the mind in the human body?	CO3	L29
22	According to the Vaiśeṣika School, how is the most powerful energy produced, and what modern invention does this principle predate?	CO3	L30
23	According to Vedic hymns and the Śatapatha Brāhmaṇa, in what natural forms does the principle of Agni manifest?	CO6	L11
24	How does the Vedic concept of Agni as the "supporter of the universe" align with modern thermodynamics?	CO4	L10
25	In the context of modern science, why is Agni addressed as the "messenger" (dyūtam) of the gods in over a hundred Ṛgvedic mantras?	CO5	L12
26	Why is it considered mathematically correct that the Vedic seers always addressed the storm deities, the Maruts, in the plural?	CO3	L13
27	How did the Vedic approach conceptualize natural forces such as fire, wind, and water?	CO3	L14
28	What is the concept of <i>Rta</i> in Vedic literature, and which deity is considered its lord?	CO1	L15

29	According to the tripartite model of the universe in Vedic texts, what are the three divisions of the universe?	CO2	L16
30	Describe the basic process of the water cycle as it is conceptualized in the R̥gveda.	CO3	L17
31	What strict ancient water protection law is stated in the Padma Purāṇa?	CO2	L18
32	How do the Purāṇas emphasize the importance of planting and protecting trees?	CO6	L19
33	What are the three core principles or components that make up the concept of <i>Yajña</i> ?	CO4	L20
34	How do the Vedic concepts of <i>nada</i> and <i>sabda</i> differ from the modern definition of sound?	CO5	L31
35	Who is revered as the Goddess of Speech, and what was her manifested purpose according to the Yajurveda mantra?	CO1	L32
36	According to the Taittiriya Upanishad, list any four of the six factors essential for precision in the oral transmission of chants.	CO2	L33
37	What are the two types of <i>Nada</i> mentioned in the text?	CO6	L34
38	According to the Vedic physiological view, where are the <i>para</i> and <i>madhyama</i> stages of speech located within the body?	CO5	L35
39	In the psychological integration of speech, what do the <i>pasyanti</i> and <i>madhyama</i> stages represent?	CO3	L36
40	What forms of life expressions does the fully manifest <i>vaikhari</i> stage encompass?	CO3	L11
41	How did the medical expertise of British doctors, such as Gabriel Boughton in 1645, benefit the East India Company politically and economically?	CO2	L12
42	Who compiled the first worthwhile map of India, and in what year was it compiled?	CO1	L12
43	What major geodetic theory and mathematical model emerged from the data collected by the Great Trigonometrical Survey of India (GTS)?	CO4	L12
44	Why was the Astronomical Observatory set up at Madras in 1790, and what was its real underlying aim according to the East India Company?	CO6	L13
45	What critical medical discovery did Ronald Ross make in 1897, and what global engineering project did his expertise later facilitate?	CO5	L24
46	Who is described as the "Father of Indian Ornithology," and what was his primary role when he was brought to India in 1841?	CO4	L14
47	Mention any two Indian traditional artists who were employed by European scientists for creating scientific drawings.	CO3	L15

48	How were Indian assistants, such as the explorer Nain Singh, typically treated by the European scientific community during the colonial era?	CO2	L16
49	Analyze the "Paradox of Necessity vs. Fear" that defined the British East India Company's policies during the "Moonshee Phase." How did this paradox ultimately lead to the end of this phase?	CO1	L17
50	Evaluate Rajendralal Mitra's sociological argument regarding the role of "Industrial Art" in 19th-century India. How was this concept intended to act as a social disruptor?	CO6	L18
51	Contrast the educational objectives and funding models of British Government Colleges with Dr. Mahendralal Sircar's vision for the Indian Association for the Cultivation of Science (IACS).	CO5	L19
52	The establishment of the Indian Institute of Science (IISc) in 1911 is described as a "colonial paradox." Synthesize the factors that contributed to this paradox and explain its significance.	CO2	L20
53	"The elite celebrated 'Science Speculation'... but ignored 'Science Application'." Evaluate this statement in the context of colonial trauma. Why did the post-independence era inherit a bias against applied science?	CO4	L21
54	Discuss the strategic significance of Sir Asutosh Mookerjee's policies at the University College of Science and Technology. How did his approach ensure academic sovereignty?	CO4	L22
55	Analyze the relationship between state intervention and agricultural expansion in Medieval India. How did the actions of rulers directly impact the empire's economy?	CO2	L23
56	Evaluate the impact of European contact on the horticultural and agricultural landscape of India during the 16th and 17th centuries.	CO1	L24
57	Compare and contrast the regional adaptations in irrigation technologies across Medieval India. How did different regions engineer solutions for their specific geographical challenges?	CO5	L25
58	Examine the role of cash crops in integrating the medieval Indian economy with global trade networks. Provide specific examples from the text.	CO2	L26
59	"By the end of the medieval period, agriculture was stabilized through a blend of administration and engineering." Assess the validity of this statement using evidence of both infrastructure and administrative systems.	CO6	L26
60	How does the Vedic concept of <i>Rta</i> (Cosmic Order) govern the environmental balance, and what are the implications of this universal law for natural forces?	CO3	L27
61	Analyze the Vedic Tripartite Model of the universe. How does the assignment of different natural forces and deities to these specific regions reflect an ancient understanding of interconnected ecosystems?	CO4	L27
62	Evaluate the practical environmental benefits of the <i>Yajña</i> ritual as described in the Yajurveda. How does this practice embody the "theory of give and take" in nature?	CO2	L28
63	Examine the Vedic classification of water sources (<i>Āpas</i>) and the ancient legal stance on water pollution. How do these concepts reflect an early understanding of the hydrological cycle?	CO1	L29
64	The Atharvaveda states, "O Earth! What on you, I dig out, let that quickly grow over." Synthesize this statement with the broader Vedic message regarding sustainable resource consumption and biodiversity.	CO6	L29
65	How does the Rgvedic description of Agni as the "supporter of the universe" (<i>dhartā, viśvasya dhāraṁ</i>) logically align with the modern Kinetic Theory of Heat and Boltzmann's energy law?	CO2	L30

66	Evaluate the scientific reasoning behind the Vedic seers addressing Agni as a "messenger" (<i>dyūtam</i>) in over 110 mantras, rather than assigning this role to other deities like Indra or the Maruts.	CO5	L26
67	Synthesize the Rgvedic concept of Agni's "seven rays" (<i>saptaraśmim</i>) with the modern understanding of polychromatic light. What does this reveal about ancient observational methods?	CO2	L34
68	Analyze how the Vedic descriptions of the Maruts (Storm Deities) implicitly encode complex principles of statics and mechanics.	CO2	L34
69	Assess the Vaiśeṣika School's ancient hypothesis regarding energy generation from contrary substances. How does this principle connect to the later technological revolution in Europe?	CO5	L35
70	Analyze the fundamental differences between the modern scientific model of speech production (as described by Honda) and the Vedic fourfold model.	CO1	L36
71	Evaluate how the ancient Vedic tradition maintained the precision of orally transmitted chants over thousands of years, and explain its significance in relation to the overarching "sound field."	CO2	L36
72	Synthesize the physiological locations and psychological functions of the four stages of speech (<i>para</i> , <i>pasyanti</i> , <i>madhyama</i> , and <i>vaikhari</i>) as described in Vedic literature.	CO2	L35
73	How do the Purāṇas emphasize the importance of planting and protecting trees?	CO6	L31
74	How do the Vedic concepts of <i>nada</i> and <i>sabda</i> differ from the modern definition of sound?	CO6	L30

Assignment 3	Topic: Indian response to new scientific knowledge	Date of Assignment1: 25.04.2026	Date of Submission: 04.05.2026
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Note:

1. assignment carries weightage of **20 marks out of 100**
2. Course outcome CO1 through CO6 was covered.

Course Outcomes	CO1	To provide a comprehensive overview of the historical development of scientific thought and technological practices in ancient India, from the Vedic period to the medieval era.
	CO2	To analyse ancient Indian contributions to fundamental sciences, Mathematics (<i>Sulba – Sutras</i>), Astronomy (<i>Jyotisha</i>), and Physics.
	CO3	To evaluate technological advancements in fields such as Metallurgy, Civil Engineering, and Architecture (<i>Vastu Shastra</i>).
	CO4	To examine the evolution of medicinal systems, specifically the theories and surgical practices found in the foundational texts of Ayurveda
	CO5	To foster a critical understanding of how ancient Indian scientific knowledge was documented in Sanskrit texts and its subsequent influence on global scientific thought through translation and trade.
	CO6	To investigate the Vedic and Post-Vedic understanding of heat transfer, latent heat, and the role of Agni in the phase transitions of matter (solid to liquid to gas).